ALLEN

(Academic Session: 2026–2027)

TALLENTEX PRACTICE QUESTIONS

CLASS 10

PHYSICS

Your Hard Work Leads to Strong Foundation

Practice Questions

1. Sound waves travel at 350 m/s through warm air and at 3500 m/s through brass. The wavelength of a 700 Hz sound wave as it enters brass from warm air :

(1) decreases by a factor 10 (2) increases by a factor 20

(3) increases by a factor 10 (4) decreases by a factor 20

2. A balloon is rising vertically upwards at a velocity of 10 m/s. When it is at a height of 45 m from the ground, a parachutist bails out from it. After 3 seconds he opens his parachute and decelerates at a constant rate of 5 m/s^2. What was the height of the parachutist above the ground when he opened his parachute ? (Take g = 10 m/s^2)

(1) 15 m (2) 30 m (3) 45 m (4) 60 m

3. A 50 kg skater at rest on a frictionless rink throws a 2 kg ball, giving the ball a velocity of 10 m/s. Which statement describes the skater's subsequent motion?

(1) 0.4 m/s in the same direction as the ball's motion

(2) 0.4 m/s in the opposite direction of the ball's motion

(3) 2 m/s in the same direction as the ball's motion

(4) 2 m/s in the opposite direction of the ball's motion

4. Optical density of a medium depends on

(1) density of the medium (2) thickness of the medium

(3) velocity of light in that medium (4) None of these

5. A ball having initial velocity v is dropped from a height of 10 m. It loses 50% of its energy on striking the ground and rises to the same height after collision. The value of v is (Take g = 9.8 m/s^2).

(1) 7 m/s (2) 14 m/s

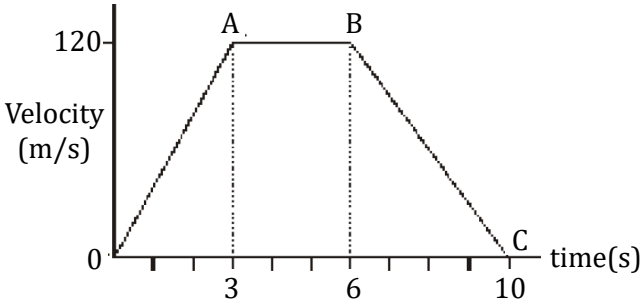
(3) 35 m/s (4) event seems impossible

6. The table below lists the masses and distances of several asteroids that orbit the Sun. For which asteroid is the gravitational force between the Sun and asteroid strongest?

Mass and Distance Data for Asteroids			
Asteroid	**Asteroid mass (kg)**	**Sun mass (kg)**	**Mean asteroid distance from Sun (km)**
Psyche	2×10^{19}	2.0×10^{30}	4.37×10^{8}
Ida	4×10^{16}	2.0×10^{30}	4.28×10^{8}
Pallas	3.2×10^{20}	2.0×10^{30}	4.15×10^{8}
Ceres	8.8×10^{20}	2.0×10^{30}	4.14×10^{8}

(1) Psyche (2) Ida (3) Pallas (4) Ceres

7. The velocity-time graph of an object of mass m = 50 g is shown in figure. Observe the graph carefully and calculate the force acting on the object in time intervals (i) 0-3 s and (ii) 6-10s



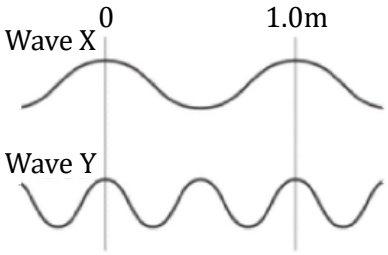
(1) (i) 40 N and (ii) 30 N (2) (i) 2 N and (ii) 1.5 N

(3) (i) 200 N and (ii) -1500 N (4) (i) 2 N and (ii) -1.5 N

9. A body, having kinetic energy k, moving on a rough horizontal surface, is stopped in a distance x. The force of friction exerted on the body is

(1) $\frac{k}{x}$ (2) $\frac{\sqrt{k}}{x}$ (3) $\frac{k}{\sqrt{x}}$ (4) kx

10. Which of the following correctly compares the two waves?



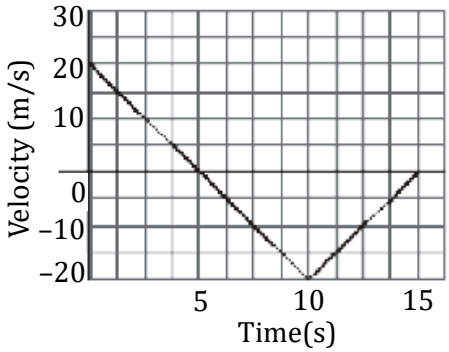
(1) Wave X has half the amplitude of wave Y.

(2) Wave X has twice the amplitude of wave Y.

(3) Wave X has a lower frequency and longer wavelength than wave Y.

(4) Wave X has a higher frequency and shorter wavelength than wave Y.

11. The graph below represents a particle moving along a straight line. What is the total distance travelled by the particle from t = 0 to t = 10 s ?



(1) 0 m (2) 50 m (3) 100 m (4) 200 m

12. As compared to the original position, the stars appear_____ then its actual position as viewed from earth's surface.

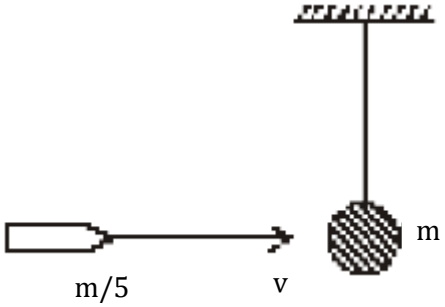
(1) to the left (2) to the right (3) higher (4) lower

13. A curtain hanging in a moving aeroplane hits the face of the passenger sitting behind it. At this particular time, the aeroplane is

(1) accelerating forward (2) accelerating backward

(3) moving forward with uniform speed (4) being braked upon

14. A bullet of mass m moving horizontally with velocity v collides against a pendulum bob of mass m and gets embedded in it. The maximum height reached by the pendulum-bullet system is



(1) $v^2/12\ g$ (2) $v^2/8\ g$ (3) $v^2/36\ g$ (4) $v^2/72\ g$

15. The diagram below shows a person holding a ball standing at three different places on Earth. If the person drops the ball, gravity will make it fall. Which of the following diagrams best shows the direction the dropped ball will fall at the three different positions ?



(1) 

(2) 

(3) 

(4) 

16. When light travels from one medium to another, which are separated by a sharp boundary, the characteristic which does not change is

(1) velocity (2) wavelength (3) frequency (4) amplitude

17. The force necessary to stop a hammer having a momentum of 25 N-s in 0.05 seconds is

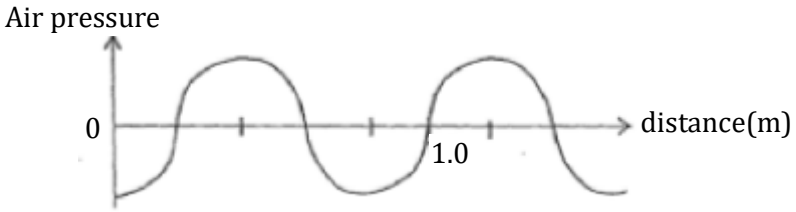
(1) 25 N (2) 50 N (3) 1.25 N (4) 500 N

18. A rope ladder with a length l carrying a man with a mass m at its end is attached to the basket of a balloon with a mass M. The entire system is in equilibrium with the air. As the man climbs up the ladder into the balloon, the balloon descends by a height h. Then the potential energy of the man

(1) increases by mg $(l - h)^4$ (2) increases by mgl

(3) increases by mgh (4) increases by mg (2l - h)

19. The diagram shows the variation of air pressure from an alarm siren at particular instant. The speed of sound in the air is 340 m/s.



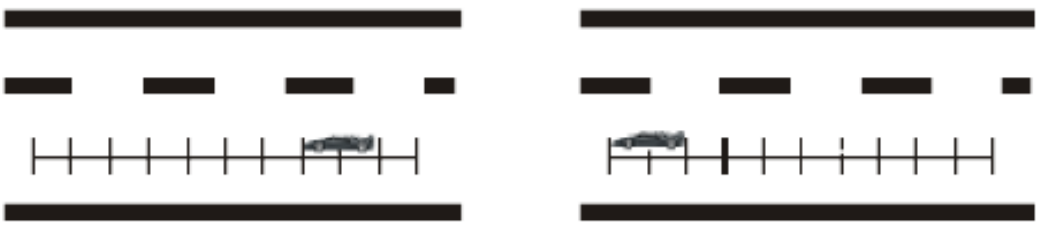
What is the period of alarm siren?

(1) 0.98 ms (2) 1.2 ms (3) 2.35 ms (4) 2.9 ms

20. Rays from Sun converge at point 15 cm in front of concave mirror. Where should an object be placed so that produce an image of magnification 2 which is real and inverted ?

(1) 15 cm (2) 30 cm (3) 7.5 cm (4) 22.5 cm

21. Speed cameras are used to detect motorists who break the speed limit. A number of lines 2 m apart are painted on the road. As a speeding car crosses the painted lines, the camera takes two photographs, 0.5 second(s) apart.



Photograph 1 Photograph 2

The speed of the car in the photographs is

(1) 10 m/s (2) 14 m/s (3) 70 m/s (4) 28 m/s

22. A child is stuck on a frictionless horizontal surface and cannot exert any horizontal force by pushing against the surface. How can he get off?

(1) By running (2) By rolling

(3) By jumping (4) By spitting or coughing

23. Two bodies of equal weight are kept at heights 'h' and '3h' respectively. The ratio of their PE is

(1) 1 : 3 (2) 3 : 1 (3) 1 : 9 (4) 9 : 1

24. Danger signals are red in colour because

(1) Red lights look attractive.

(2) Red coloured bulbs consume least power.

(3) Red coloured light is scattered the most by fog or smoke particles, so visible even from long distance.

(4) Red coloured light is scattered the least by fog or smoke particles, so visible even from long distance.

25. The table below shows the mass and the weight of a certain object on Earth.

Mass (kg)	Weight (N)
6.0	60

What are the approximate mass and approximate weight of the same object on the Moon?

(1)

Mass (kg)	Weight (N)
1.0	10

(2)

Mass (kg)	Weight (N)
6.0	60

(3)

Mass (kg)	Weight (N)
6.0	360

(4)

Mass (kg)	Weight (N)
36.0	360

26. An isolated system consists of two bodies on which no external forces act. The two bodies collide with each other and stick together on impact.

	Total kinetic energy before and after collision	Total momentum before and after collision
A	different	different
B	different	the same
C	the same	different
D	the same	the same

Which row correctly compares the total kinetic energy and the total momentum of the bodies before and after the collision?

(1) A (2) B (3) C (4) D

27. Four students were asked to classify the activities of the people in the picture below as examples of either potential or kinetic energy.

Which student correctly classified the activities?

(1)

Student 1	
Activity Observed	**Classification of Activity**
Girl swimming laps	Potential energy
Boy on diving board	Kinetic energy
Girl hitting volleyball	Potential energy
Boy holding volleyball	Kinetic energy

(2)

Student 2	
Activity Observed	**Classification of Activity**
Girl swimming laps	Potential energy
Boy on diving board	Potential energy
Girl hitting volleyball	Kinetic energy
Boy holding volleyball	Kinetic energy

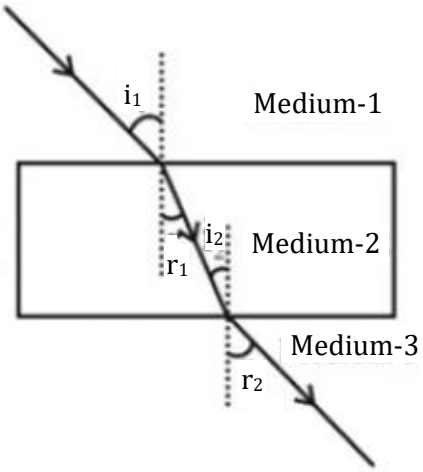
(3)

Student 3	
Activity Observed	**Classification of Activity**
Girl swimming laps	Kinetic energy
Boy on diving board	Kinetic energy
Girl hitting volleyball	Potential energy
Boy holding volleyball	Potential energy

(4)

Student 3	
Activity Observed	**Classification of Activity**
Girl swimming laps	Kinetic energy
Boy on diving board	Potential energy
Girl hitting volleyball	Kinetic energy
Boy holding volleyball	Potential energy

28.



In the above diagram light is travelling through different media. It is observed that ? i = ?r but ?i < ?r . Which of the following statement would be correct?

(1) Medium 1 is denser than medium 2 but it's density is equal to medium 3.

(2) Medium 2 is the rarest medium.

(3) Medium 3 is denser than medium 2.

(4) Medium 1 and 3 are essentially the same medium, but medium 2 is denser than 1 and 3.

29. Which of the following is the characteristic of sound wave?

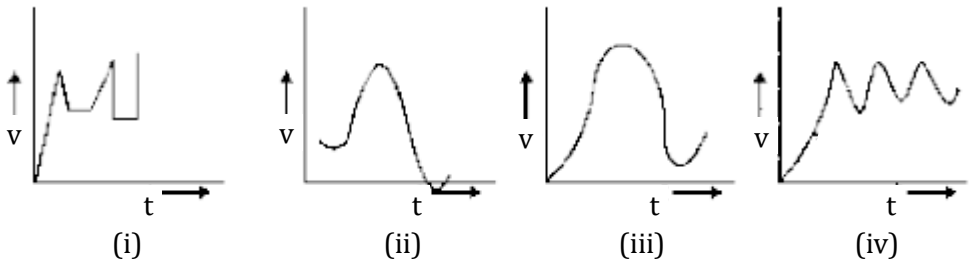
(1) It is non mechanical and longitudinal wave

(2) It is mechanical and transverse wave

(3) It is non mechanical and transverse wave

(4) It is mechanical and longitudinal wave

30. The following figures show velocity v versus time t curves. But only some of these can be realised in practice. These are



(1) (i), (ii) and (iv) only

(2) (i), (ii) and (iii) only

(3) (ii) and (iv) only

(4) all

31. In drawing 1 below when the boatman pushes against the bank with one oar, the boat moves in the opposite direction. In drawing 2, when he rows the boat, the oars push against the water in one direction and the boat moves in the opposite direction.



Use the above information to answer the question below an inflated balloon is fastened to a car as shown below. The opening of the balloon is at the back of the car. What will happen when air escapes from the balloon? The car will...



(1) move backward (2) move sideways

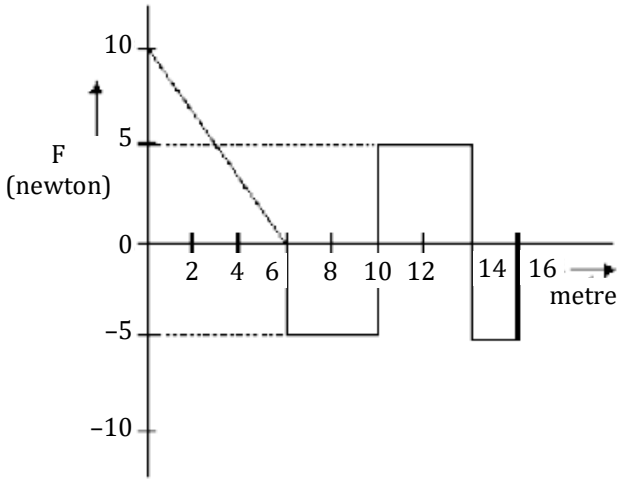
(3) not move (4) move forward

32. A child is standing in front of a magic mirror. She finds the image of her head bigger, the middle portion of her body of the same size and that of the legs smaller. The following is the order of combinations for the magic mirror from the top

(1) Plane, convex and concave (2) Convex, concave and plane

(3) Concave, plane and convex (4) Convex, plane and concave

33. A particle is acted upon by a force F which varies with position x as shown in figure. If the particle at x = 0 has kinetic energy of 25 J, then the kinetic energy of the particle at x = 16 m is



(1) 45J (2) 30J (3) 70J (4) 135J

34. If suddenly the gravitational force of attraction between earth and a satellite revolving around it becomes zero, then the satellite will

(1) continue to move in its orbit with same velocity

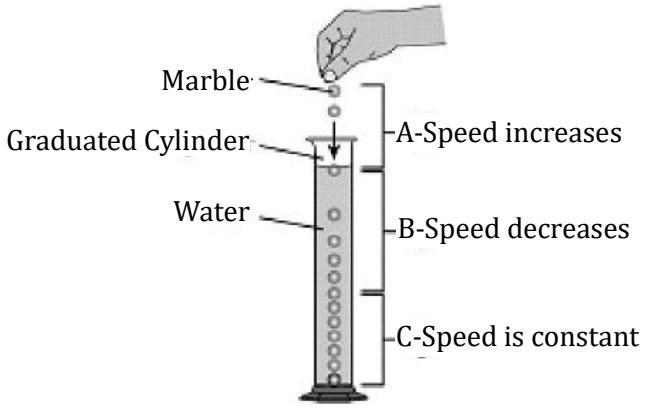
(2) move tangentially to the original orbit with the same velocity

(3) become stationary in its orbit

(4) move towards the earth.

35. A student used a video camera to record another student dropping a marble through water in a graduated cylinder. The students watched the video in slow motion and made the observations shown below.

Marble Drop Experiment



During which part or parts of the marble's fall did the marble experience unbalanced forces?

(1) Part A only (2) Parts A and B only

(3) Part C only (4) Parts B and C only

36. Which of the following statements is correct?

(1) A person with myopia can see nearby objects clearly.

(2) A person with hypermetropia can see nearby objects clearly.

(3) A person with myopia can see distant objects clearly.

(4) A person with hypermetropia cannot see distant objects clearly.

37. **Assertion :** An observer standing on the bank of sea finds that 54 waves are reaching the bank per minute.

If the wavelength of waves is 10 m, then their speed will be 9 m/s.

Reason : This follows from v = f

(1) If both Assertion and Reason are true and Reason is correct explanation of the Assertion.

(2) If both Assertion and Reason are true but Reason is not correct explanation of the Assertion.

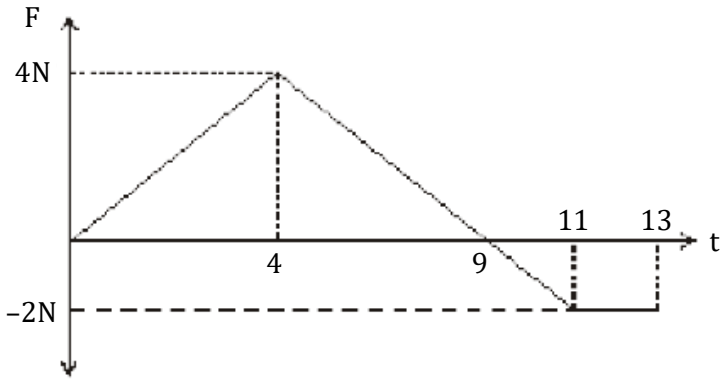
(3) If Assertion is true, but the Reason is false.

(4) If Assertion is false, but the Reason is true.

38. A ball is released from height h and another from 2h. The ratio of time taken by the two balls to reach the ground is :

(1) $1:\sqrt{2}$ (2) $\sqrt{2}:1$ (3) 2 : 1 (4) 1 : 2

39. In the graph below is given the variation of force with time. Find out the net change in momentum of the object.



9

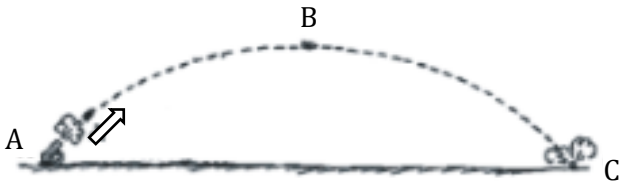
(1) 24 kg m/s (2) 12 kg m/s (3) 6 kg m/s (4) 3 kg m/s

40. Which of the following phenomena involves total internal reflection of light?

(1) Sparkling of diamond (2) Rainbow formation

(3) Optical fibre communication (4) All of these

41. A bullet is fired from a cannon as shown in the figure. What happens when bullet reaches from point A to point B ?



(1) Kinetic energy increases, potential energy decreases.

(2) Potential energy increases, kinetic energy remains constant.

(3) Potential energy increases, mechanical energy remains constant.

(4) Kinetic energy decreases, potential energy remains constant.

42. The gravitational force between two object does not depends on

(1) Sum of masses (2) Product of masses

(3) Gravitational constant (4) Distance between the masses

43. Which of the following statement is incorrect?

(1) Newton's III law is valid for non-contact forces

(2) Normal force always acts perpendicular to surfaces in contact

(3) Friction is a dissipative force

(4) If a person is walking in south, the friction on his feet acts in north

44. Magnification (m) provided by a convex mirror can be

(1) -1 (2) Zero (3) +1 (4) +0.5

45. Two men with their weights in the ratio 5 : 3 run up a staircase in times in the ratio 11 : 9. Then the ratio of power of first to that of second is-

(1) $\frac{15}{11}$ (2) $\frac{11}{15}$ (3) $\frac{11}{9}$ (4) $\frac{9}{11}$

46. In SONAR, we use

(1) Ultrasonic waves (2) Infrasonic waves (3) Radio waves (4) None of these

47. If a car covers of the total distance with v_1 speed and remaining distance with v_2, then average speed is-

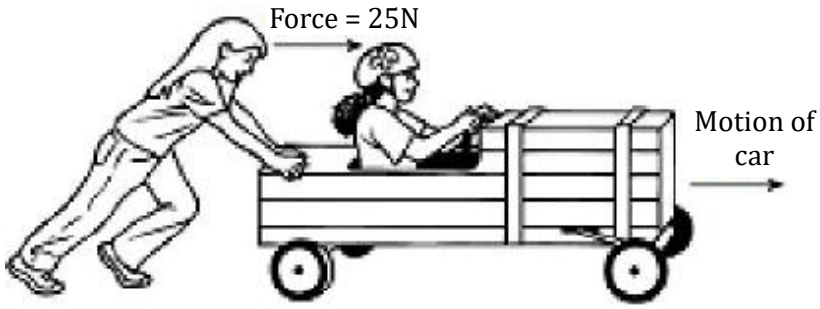
(1) $\frac{v_1 v_2}{v_1 + v_2}$ (2) $2(v_1 + v_2)$ (3) $\frac{v_1 v_2}{v_1 - v_2}$ (4) $\frac{2v_1 v_2}{v_1 + v_2}$

48. When objects at different distances are seen by the eye, which of the following remains constant ?

(1) Focal length of eye lens (2) Object distance from eye lens

(3) Radii of curvature of eye lens (4) Image distance from eye lens

49. The diagram shows a homemade car being pushed with a force of 25 N.



The force causes the car to move at a constant speed of 3 m/s. What will happen if the force is changed to 35 N?

(1) The car will move at a constant speed of 13 m/s.

(2) The speed of the car will not change.

(3) The speed of the car will increase.

(4) The speed of the car will decrease to 1 m/s.

50. A boy is sitting on a swing which can go upto a maximum height of 5 m from the ground. When it returns from maximum height to its lowest position which is at height 2 m above the ground, its velocity at lowest position is (approximately) ($g = 10\ m/s^2$)

(1) 9.8 m/s (2) 7.7 m/s (3) 6.26 m/s (4) 8.2 m/s

51. If you were asked to carry a 10 kg mass piece on each of the planets, select the planets in order, from the one where you would find it lightest to the one where it would be the heaviest

(1) Mercury - Venus - Earth - Mars - Jupiter

(2) Jupiter - Mars - Earth - Venus - Mercury

(3) Mercury - Mars - Venus - Earth - Jupiter

(4) Mars - Mercury - Venus - Earth - Jupiter

52. An object is placed at a distance of 10 cm from a convex mirror of focal length 15 cm. Find the position and nature of the image.

(1) Position: 6 cm behind the mirror; Nature: virtual and erect

(2) Position: 6 cm in front of the mirror; Nature: virtual and erect

(3) Position: 6 cm behind the mirror; Nature: real and inverted

(4) Position: 6 cm in front of the mirror; Nature: real and inverted

53. When balanced forces acts on a body, the body :

(1) must continue its state of rest, if already at rest

(2) must continue moving with uniform velocity, if already in uniform motion

(3) must experience some acceleration

(4) both (1) and (2)

54. If the magnitude of momentum of a certain body be increased by 50%, its kinetic energy will increases by:

(1) 125% (2) 150% (3) 50% (4) 62.5%

55. The frequency of a wave travelling at a speed of 500 m/s is 25 Hz. Its time period will be_____.

(1) 20 s (2) 25 s (3) 0.05 s (4) 0.04 s

56. A body stands in front of a mirror at a distance of 30 cm away from it. He sees his erect image whose height is 1/5th of his real height. The mirror he is using is

(1) Plane mirror (2) Convex mirror

(3) Concave mirror (4) Plane-concave mirror

57. A body is dropped from the roof of a multi storied building it passes the ceiling of the 20th storey at a speed of 20 m/sec if the height of each storey is 5m, then the number of storeys in the building .

(1) 24 (2) 20 (3) 25 (4) 30

58. A force of 5 N acts on a body of weight 10 N, then the acceleration produced by it is (take g = 10 m/s^2):

(1) 2 m/s^2 (2) 5 m/s^2 (3) 0.5 m/s^2 (4) 50 m/s^2

59. Which of the following is not a unit of work :

(1) newton × metre (2) joule

(3) $\frac{\text{kilogram}\times(\text{metre})^2}{(\text{second})^3}$ (4) $\frac{\text{kilogram}\times(\text{metre})^2}{(\text{second})^2}$

60. Twinkling of stars is due to atmospheric

(1) dispersion of light by water droplets

(2) refraction of light by different layers of varying refractive indices

(3) scattering of light by dust particles

(4) internal reflection of light by clouds

61. A man suffering from myopia can read a book placed at 10 cm distance. For reading the book at a distance of 60 cm with relaxed vision, focal length of the lens required will be

(1) 45 cm (2) 20 cm (3) 12 cm (4) 30 cm

62. The distance between three consecutive troughs in a wave produced in a string is 4 cm. If 2.5 wave cycles pass through any point in a second, the wave in the string has velocity

(1) 10 cm/s (2) 0.16 cm/s (3) 0.8 cm/s (4) 5 cm/s

63. The velocity of a body moving with a uniform retardation of 5 m/s^2 is 100 m/s at an instant. Its velocity after 7 second is

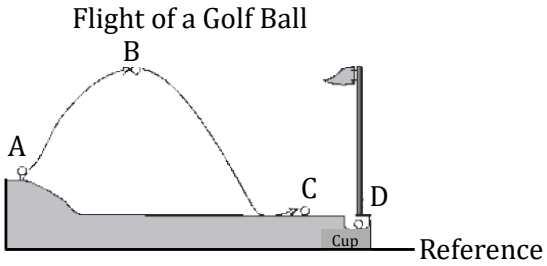
(1) 135 m/s (2) 100 m/s (3) 65 m/s (4) Zero

64. A child is stuck on a frictionless horizontal surface and cannot exert any horizontal force by pushing against the surface. How can he get off?

(1) By running (2) By rolling

(3) By jumping (4) By spitting or coughing

65. Eddie hits a golf ball from the tee into the cup. At point A, the ball is stationary on the tee. At point B, the ball is at its highest point in the air. At point C, the ball is rolling on the ground. At point D, the ball is stopped in the cup. The diagram below shows the path the ball traveled from the tee to the cup.



Choose the correct option

At A	From A to B	From B to C	At D
(1) Only PE	Both KE + PE	Both KE + PE	Only PE
(2) Only KE	Both KE + PE	Both KE + PE	Only PE
(3) Only PE	Only PE	Only KE	Only KE
(4) Only KE	Only PE	Only KE	Only PE

66. The refractive index of air w.r.t. glass is 2/3. The refractive index of diamond w.r.t. air is 12/5. Then, the refractive index of glass w.r.t. diamond will be

(1) 5/8 (2) 8/9 (3) 5/18 (4) 18/5

67. An iron sphere of mass 20 kg has same diameter as an aluminium sphere of mass 12 kg. Both spheres are dropped from towers of same height. When they are 10 meter above the ground, they have the same (neglect air resistance)

(1) acceleration (2) velocity (3) kinetic energy (4) both (1) and (2)

68. A force of 980 N acts on a body for 0.1 seconds. Calculate the change in momentum of the body.
(1) 98 N s (2) 9.8 N s (3) 0.98 N s (4) 980 N s

69. Two men with their weights in the ratio 5 : 3 run up a staircase in times in the ratio 11 : 9. Then the ratio of power of first to that of second is-
(1) $\frac{15}{11}$ (2) $\frac{11}{15}$ (3) $\frac{11}{9}$ (4) $\frac{9}{11}$

70. Magnification of mirror is +2. Identify type of mirror and position of object.
(1) Concave mirror, object is beyond C (2) Concave mirror, object is at P
(3) Concave mirror, object is between P & F (4) Convex mirror object is in front of mirror

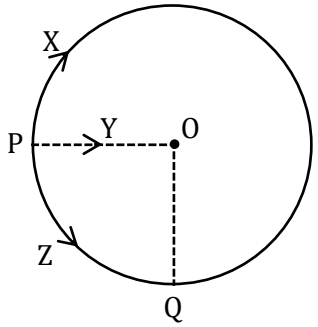
71. Which of the following statement is correct ?
(1) If force is along velocity, then it can change the direction of motion.
(2) If force is perpendicular to velocity, then it can change the magnitude of velocity.
(3) If force is in the opposite direction of velocity, then it can not change the direction of motion.
(4) If force is perpendicular to velocity then it can change direction of velocity.

72. Two cars with different types of tyres are decelerating in summer on a straight dry road from a velocity v_0 = 10 km-h^{-1}. Initially, they are moving next to each other and start braking at the same time. When the car with summer tyres comes to the rest the other one, with winter tyres, is still moving with a velocity v_1 = 3.7 km-h^{-1} and stops after 6 more meters. What is the fraction of the horizontal deceleration of the car with winter tyres to the horizontal deceleration of the other car?
(1) 0.37 (2) 0.63 (3) 0.67 (4) 0.74

73. The weight of an object in the coal mine, sea level and at the top of the mountain are W_1 W_2 and W_3 respectively, then
(1) $W_1 < W_2 > W_3$ (2) $W_1 = W_2 = W_3$ (3) $W_1 < W_2 < W_3$ (4) $W_1 > W_2 > W_3$

74. A mass of 2 kg moving initially at 10 m/s accelerates and travels for 4 sec with an acceleration of 1.5 m/s^2. What is the percentage gain of momentum?
(1) 40% (2) 50% (3) 60% (4) 80%

75. Three particles x, y and z are situated at point P on the circular path of radius 10 m. All three particles move along different paths and reach point Q as shown in figure. Then the ratio of distance travelled by particles x and y is :

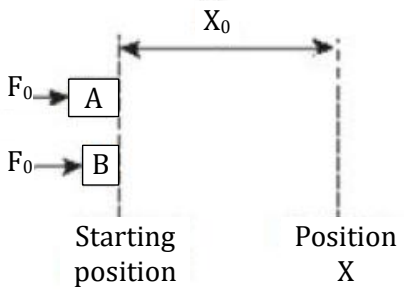


(1) $\frac{3}{4}$ (2) $\frac{1}{3}$ (3) $\frac{3\pi}{4}$ (4) $\frac{\pi}{3}$

76. A source of sound produces waves of wavelength 0.8 m in air. The same source of sound produces wave of wavelength 4 m in water. If the velocity of sound in air is 332 m/s. Find the velocity of sound in water?

(1) 2660 m/s (2) 1660 m/s (3) 830 m/s (4) zero

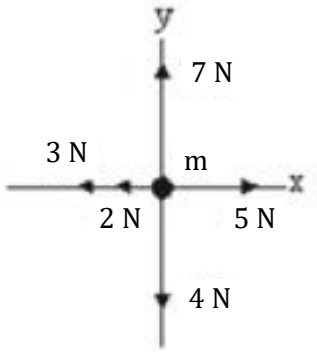
77. Carts A and B are initially at rest on a frictionless, horizontal surface. A constant force F_0 is applied to each cart as it travels from its initial position. The mass of cart A is more than the mass of cart B. Consider the kinetic energy (E), and momentum (p), of the boxes at position X, a distance X_0 from the initial position. Which statement below is correct?



(1) $E_A < E_B$, $p_A < p_B$ (2) $E_A < E_B$, $p_A = p_B$

(3) $E_A = E_B$, $p_A = p_B$ (4) $E_A = E_B$, $p_A > P_b$

78. Figure below shows the situation in which an object of mass 'm' is pulled by several forces across a frictionless floor, as seen from overhead.



The velocity versus time graph for the body in consideration will be

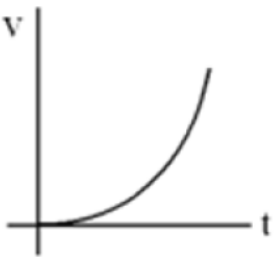
(1)



(2)



(3)



(4)



79. **Assertion:** A stone tied with a piece of a thread describing a circular path, moves in a straight line on being released.

Reason: Along the circular path direction of motion remains the same at every point.

(1) both assertion and reason are true and the reason is the correct explanation of the assertion.

(2) both assertion and reason are true but reason is not the correct explanation of the assertion.

(3) assertion is true but reason is false.

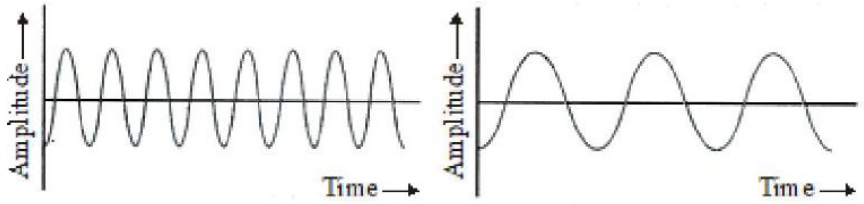
(4) assertion is false but reason is true

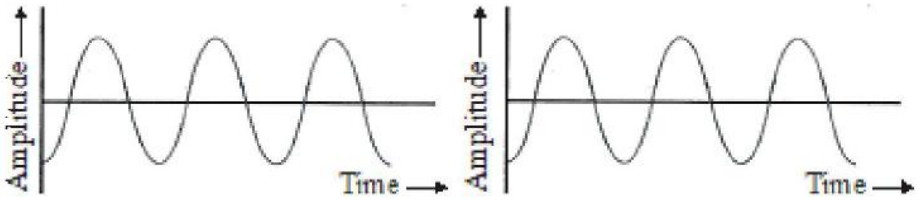
80. A body of mass 15 kg is thrown vertically upwards with an initial velocity of 40 m/s. Assuming no energy losses due to air resistance, what will be its kinetic energy when it reaches at maximum height?

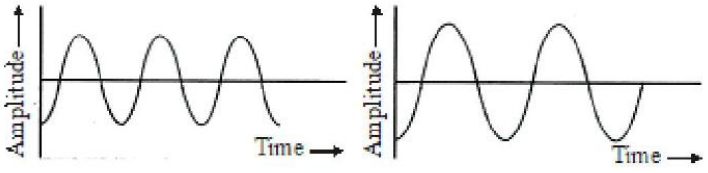
(1) 0 J (2) 10 J (3) -10 J (4) 100 J

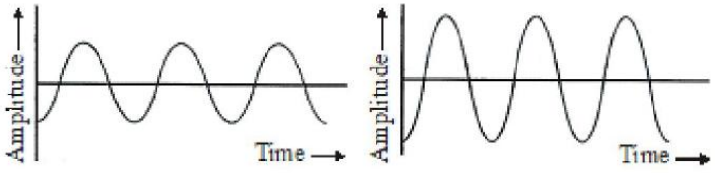
(1) A (2) A (3) A (4) A

81. Which of the below given diagrams shows two sound waves having the same amplitude but different frequencies?

(1) 

(2) 

(3) 

(4) 

82. Which of the following is an incorrect statement?

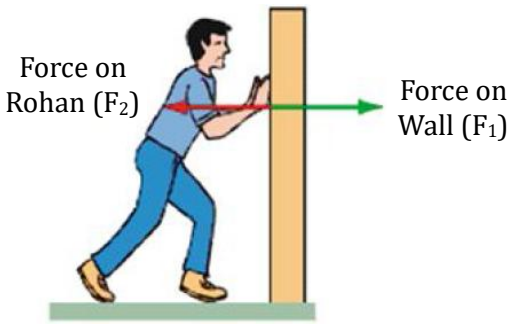
(1) Gravitational force is only attractive in nature.

(2) Gravitational force acts on the line joining the two centers of masses.

(3) Gravitational force cannot be repulsive in nature.

(4) Gravitational force does not obey inverse square law.

83. Rohan is pushing a wall harder as shown in the figure. Choose the correct statement from the following.



(1) F_1 and F_2 cancel each other.

(2) F_1 and F_2 are equal in magnitude and act in same direction.

(3) F_1 and F_2 act on the same bodies.

(4) F_1 and F_2 are equal in magnitude and act in opposite direction.

84. In a daring midnight operation, pilots of the Indian Air Force (IAF) land with a velocity of 75 m/s and comes to rest, after covering a runway of 1000 m. The retardation is

(1) -2.8125 ms^{-2} (2) -1.825 ms^{-2} (3) -4.8125 ms^{-2} (4) -6.1825 ms^{-2}

85. A full length image of a distant tall building can definitely be seen by using

(1) a concave mirror (2) a convex mirror

(3) a plane mirror (4) both concave as well as plane mirror

86. When a body is kept below the surface of Earth, it is acted upon by various forces of attraction then,

(1) acceleration due to gravity on body increases.

(2) acceleration due to gravity on body decreases.

(3) acceleration due to gravity on body remains constant.

(4) acceleration due to gravity on body may increases or decreases.

87. Minakshee wants to displace a wooden block from point A to point B along the surface of a table as shown. In doing so;



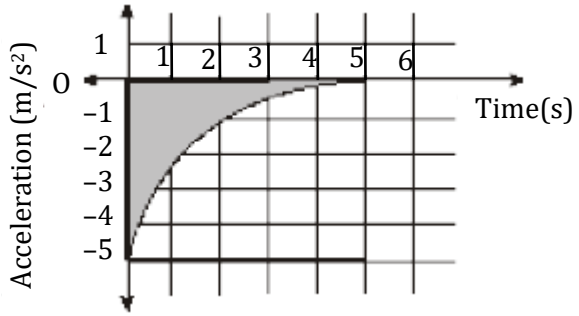
(1) Work done by applied force of Minakshee is positive.

(2) Work done by applied force of Minakshee is negative.

(3) Work done by applied force of Minakshee is zero.

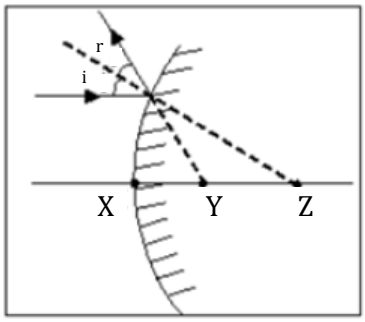
(4) Work done by applied force of Minakshee is equal to (distance between points A and B).

88. Rohan stopped his car to save a dog. His acceleration time graph is shown here. What is the force experienced by Rohan, if mass of Rohan is 50 kg. (Assume arc as quarter of a circle.)
[Hint:- Area of shaded region = (Area of square) - (Area of quarter of circle)]



(1) $250\left(1-\frac{\pi}{4}\right)$N (2) $25\left(1-\frac{\pi}{4}\right)$N (3) $125\left(1+\frac{\pi}{4}\right)$N (4) $150\left(1+\frac{\pi}{4}\right)$N

89. Identify the points X, Y & Z in the figure given below.



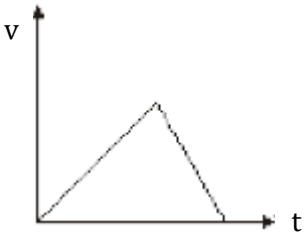
(1) X pole, Y focus, Z center of curvature

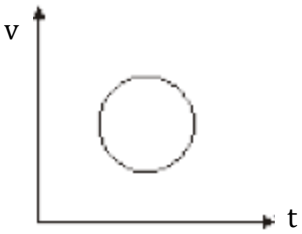
(2) X pole, Y focal length, Z radius of curvature

(3) X optical center, Y focus, Z centre of curvature

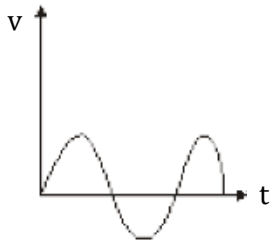
(4) X centre of curvature , Y focus, Z radius of curvature

90. Which one of the following curves is not possible for a body in motion?

(1) 

(2) 

(3) 

(4) 

91. Stone of mass 2 kg is kept at a height 20 m from the ground. When it is released after sometime its kinetic energy becomes 250 J. What will be the potential energy of the body at that instant?

(1) 150 J (2) 200 J (3) 250 J (4) 400 J

92. The far point of a myopic person is 75 cm in front of the eye. The nature and power of the lens required to correct the problem, will be

(1) convex lens, - 1.33 D (2) concave lens, - 1.33 D

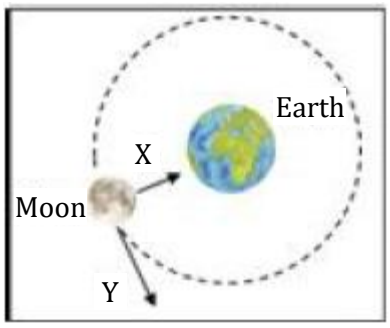
(3) concave lens, + 1.33 D (4) convex lens, + 1.33 D

93. The distance travelled by a sound wave per unit time is

(1) wave length (2) speed of sound

(3) frequency (4) time period

94. Identify the directions X and Y in the figure below



(1) X Velocity, Y Centripetal Force

(2) X Centripetal Force, Y Velocity

(3) X Velocity, Y Centrifugal Force

(4) X Centrifugal Force, Y Centripetal Force

95. If a train is moving towards east with a constant speed 20 m/s then

(1) no force is acting on the train (2) net force is acting towards east

(3) net force is acting towards west (4) resultant force on the train is zero.

96. The gravitational force between two objects is F. If masses of both the objects are halved without altering the distance between them, then the gravitational force would become

(1) F/4 (2) F/2 (3) F (4) 2F

97. If V_r is velocity of sound in a rainy day and V_d is velocity of sound in a dry day through air then

(1) $V_d = V_r$ (2) $V_r = V_d$ (3) $V_r = 2V_d$ (4) $V_r = V_d$

98. A person cannot see objects distinctly at a distance less than 2m. The power of the spectacles that he should use in order to read a paper at a distance of 0.25m is.

(1) +4D (2) +4.5D (3) +3.5D (4) -3D

99. A roller coaster car is released from the top of a hill, and it travels down the hill and through various loops and curves. Assuming no energy losses due to friction or air resistance, what can you say about the total mechanical energy of the roller coaster car at different points along the track?

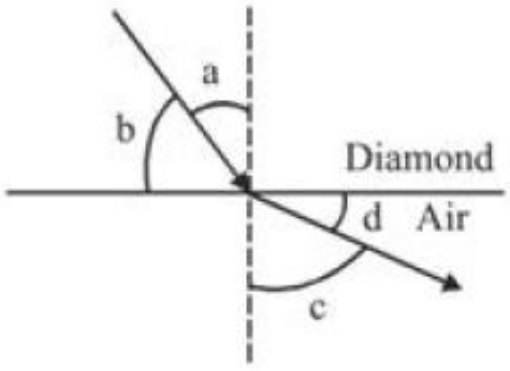
(1) The total mechanical energy is constant at all points along the track.

(2) The total mechanical energy decreases as the car goes down the hill and increases as it goes up.

(3) The total mechanical energy increases as the car goes down the hill and decreases as it goes up.

(4) The total mechanical energy of the car is zero at the top of the hill.

100. The refractive index of diamond is 2.42. Which of the following is correct for the situation shown?



(1) (sin a)/(sin b) = 2.42 (2) (sin a)/(sin c) = 2.42

(3) (cos a)/(cos c) = 2.42 (4) (sin c)/(sin a) = 2.42

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	2	2	3	2	4	4	3	1	3	3	3	1	4	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	4	1	4	4	4	4	1	4	1	2	4	4	4	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	4	3	1	2	2	1	1	1	2	4	3	1	4	4	1
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	1	4	4	3	2	3	1	4	1	4	2	1	2	3	2
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	3	4	3	4	1	1	4	1	1	3	4	2	1	3	3
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	2	4	1	4	1	1	4	4	1	2	2	1	1	1	2
Question	91	92	93	94	95	96	97	98	99	100					
Answer	1	2	2	2	4	1	4	3	1	4					